Interactive Concept Network Enhanced Transformer for Remote Sensing Image Captioning

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***Abstract*— Remote sensing image captioning plays an important role in advancing remote sensing image understanding with natural language generation. However, it is difficult to generate accurate semantic descriptions of crucial objects and their relationships, due to large coverage and abundant information in remote sensing images. To address these issues, this article proposes a novel interactive concept network enhanced transformer (ICNET) for remote sensing image captioning. First, multilevel visual features are extracted within a local and global feature extraction module. To comprehensively capture key objects in the local features, a concept mapping network (CMN) is constructed to project multiscale local features onto high-level semantic concepts of the objects. This allows for the integration of the relevant feature vectors in the visual feature mapping into multiple relatively independent word features, thus bridging the gap between visual features and semantic concepts. Subsequently, a global feature enhancement (GFE) module is introduced to boost the discrimination of global relationships and filter irrelevant content. Finally, to aggregate semantic concepts and global features, a transformer equipped with a concept interaction module (CIM) is designed to facilitate feature alignment and generate captions with proper categories and relationships. The experimental results on three remote sensing image captioning datasets demonstrate the superiority of the proposed method.**

***Index Terms*— Concept network, image captioning, relational description, remote sensing, transformer.**

I. Introduction

With the development of deep neural networks, remote sensing image processing has made significant progress in computer vision tasks such as object detection [1], [2], [3], change detection [4], [5], [6], and image segmentation [7], [8], [9]. However, these tasks usually produce single or multiple discrete labels to describe the main objects and provide limited information about the complex set of objects and scenes present in remote sensing images. To address this limitation, researchers turned to natural language processing [10], [11]. As a cross-modal task combining computer vision [12] and natural language processing [13], remote sensing image captioning aims to caption the visual content of remote sensing images in natural language [14]. It combines textual generation and visual understanding to facilitate the interpretation of remote sensing scenes, beyond discrete labels, to provide more comprehensive information about the topology and distribution of multiple objects and their attributes [15]. Remote sensing image captioning provides intuitive guidance for a wide range of application areas such as natural disaster assessment and land cover monitoring.

Compared with natural image captioning, describing remote sensing image is an arduous task because the generated captions should not only capture the ground features at different scales [16] and categories [17] but also their interrelated attributes and modalities [18]. The main challenges for remote sensing image captioning can be summarized in two aspects: First, due to the distinctive characteristics of remote sensing images, it is difficult to achieve effective abstraction and representation of important targets and interrelationships. Second, the lack of interaction and alignment between extracted remote sensing image features and text features leads to semantic inaccuracies in the generated sentences [19].

For solving the first challenge, common approaches usually extract features from remote sensing images through convolutional neural networks (CNNs) [20], [21] or visual transformer (VIT) [22]. In order to capture targets at different scales to prevent ignoring crucial targets [23], multiscale features are extracted by different convolutional layers of CNN [24]. However, CNN lacks the ability to model image global information, especially for the extraction and representation of global relationships. Compared to the ability of CNN to capture local objects, VIT is able to capture global relationships in an image through a self-attention mechanism. However, it is worth noting that VIT cannot effectively utilize local information, resulting in a limited ability to capture targets of scales and categories.

In contrast, the second challenge concerns the interaction of the visual features used for caption generation to the semantic of the text. In general, the visual features are encoded into high-dimensional feature vectors, and then, the caption

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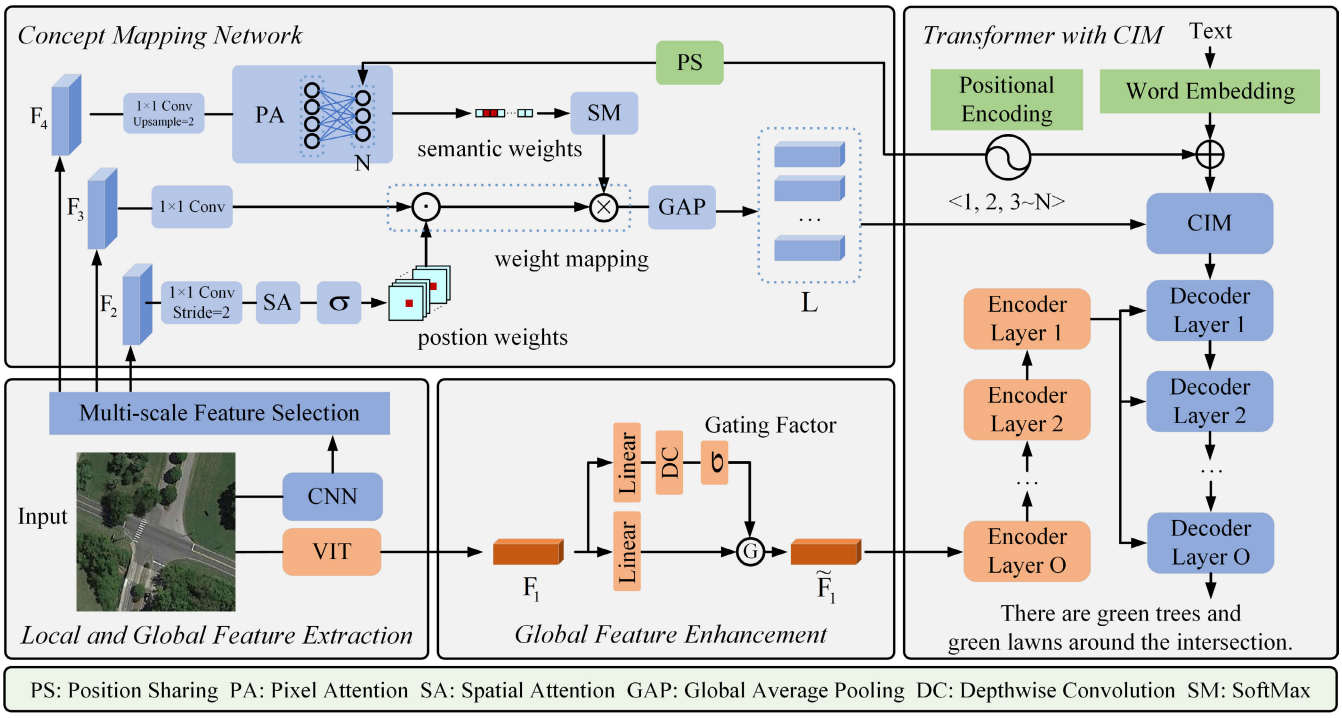


Fig. 1. Framework of the proposed ICNET. The local and global feature extraction employs a pretrained CNN and VIT image encoder to extract multiscale local features and global features from an input image, respectively. The CMN projects multiscale local features onto high-level semantic concepts. The GFE module enriches global feature representations. The transformer with CIM facilitates feature alignment and generates captions with appropriate categories and correlations.

is directly generated techniques based on recurrent neural network (RNN) [25], [26] and transformers [27], [28]. Another popular approach is by interacting visual and textual features in a cross- attention manner [29], [30], [31]. Due to the lack of good interaction ability of the model, the description of texture relationships is limited.

As a result, how to effectively utilize feature aggregation methods for different levels of image features and modeling cross-modal feature interactions of key targets and relationships to form a comprehensive visual representation is significant for remote sensing image captioning. To alleviate these problems, this article proposes a solution that combines multilevel feature aggregation and cross-modal feature interaction. To obtain comprehensive features of multiscale targets and global relationships, CNN and VIT features are jointly used for feature representation capability. A concept mapping network (CMN) is proposed to efficiently aggregate local features at different scales to generate high-level semantic concepts. A global feature enhancement (GFE) module is used to acquire more comprehensive and enhance the discriminative power of global semantic information. Furthermore, to overcome cross-modal feature interaction, the transformer's standard input or output form is followed for globally enhanced feature interactions, which improves the model's ability to discriminate global relationships. Specifically, the concept interaction module (CIM) is designed to model the interaction of semantic concepts with word vectors by integrating the relevant feature vectors in the visual feature mapping into multiple relatively independent word features.

Based on the aforementioned idea, this article proposes an interactive concept network enhanced transformer (ICNET) for remote sensing image captioning, as shown in Fig. 1. The proposed framework consists of four main components: a local and global feature extraction module, a CMN module, a GFE module, and a transformer with CIM. First, CNN is employed to extract local features of the image at different scales. The global features of the image are extracted with VIT. Then, a CMN is proposed to efficiently aggregate local features through attention in spatial and channel dimensions. A GFE module is used to acquire more comprehensive and rich feature representations. Finally, a transformer equipped with a CIM is proposed, where semantic concepts interact with word vectors through the CIM, to generate coherent and contextually relevant captions. The main contributions of this work are summarized as follows.

1) A novel ICNET is proposed to generate accurate target and relationship captions for the remote sensing image captioning task.
2) In order to enhance the generation of captions for crucial targets, the CMN module that aggregates features of different scales is proposed, and then, the CIM module is further introduced to enhance the correlation between semantic concepts and generated words and to improve the model's ability to discriminate each word in a caption.
3) The GFE is designed to enhance the global semantic information, which is then injected into the decoding network to enhance the discriminative ability of the model for relationship descriptions.

The rest of this article is organized as follows. Section II reviews related work. Section III describes the modeling and image captioning generation process. To evaluate our

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approach, we design experiments in Section IV. Finally, this article draws a conclusion in Section V.

II. Related Work

In this section, the relevant works for natural image captioning and remote sensing image captioning will be briefly introduced.

A. Natural Image Captioning

The early proposed methods for image captioning include entailed description retrieval [32], [33], [34], [35] or template filling and handcrafted natural language generation techniques [36], [37], [38], [39], [40]. Current mainstream image captioning methods for sentence generation are usually constructed by using deep learning-based generative models. In the simplest formulation, the high-level representation is extracted by activating the last layer of the CNN, which is then used as the conditional reflection element of the language model. This is the approach used in the seminal "Show and tell" paper [41], in which the output of the GoogLeNet [42] is fed into the initial hidden state of the language model. In the same year, Karpathy and Fei-Fei [43] used global features extracted from AlexNet [44] as input to a language model. In addition, Mao et al. [45] and Donahue et al. [46] injected global features extracted from the VGG network [47] at each time step of the language model. Due to the drawbacks of global representation, most of the following methods increase the granularity level of visual coding [48], [49], [50]. For example, Dai et al. [51] utilized a 2-D activation map instead of a 1-D global feature vector to introduce spatial structure directly into the language model. From the machine translation literature, a large portion of the subtitling community employs additional attentional mechanisms to endow image subtitling architectures with time-varying visual feature encoding with greater flexibility and finer granularity. Self-attention is an attentional mechanism in which each element in a set is connected to all the other elements, and a fine-grained representation of the same set of elements can be computed by residual linking. It was first introduced by Vaswani et al. [29] for machine translation and language understanding tasks, giving rise to the transformer architecture and its variants, which has dominated the natural language processing field and later on computer vision. After the emergence of global and grid features, region-based features have been the state-of-the-art choice for image captioning for many years due to their convincing performance. Recently, however, different factors have revisited which feature model is best suited for image captioning, ranging from the performance of better-trained grid features [52] to the emergence of self-attention visual coders [53] and large-scale multimodal models such as contrastive language-image pretraining (CLIP) [54]. Recent strategies include training better object detectors on large-scale data [55] or using end-to-end visual models trained from scratch [56]. Furthermore, the success of BERT-like solutions performing early fusion of images and text demonstrates the applicability of visual representations that also integrate textual information.

B. Remote Sensing Image Captioning

Compared with natural image captioning, the differences in remote sensing image captioning are mainly in data sources, content features, and language representations. First, Qu et al. [57] proposed a multimodal encoder–decoder architecture to interpret high-resolution remotely sensed images at the semantic level. In their model, features extracted from CNNs are combined with captions from RNN to generate final sentences. They demonstrate the effectiveness of their architecture on their proposed dataset, UCM-captions, and Sydney-captions. Shi and Zou [58] proposed a method to address the different semantic challenges of the same ground elements in remote sensing images at different geographical scales. They introduced a fully convolutional network (FCN) to generate comprehensive and robust sentences for remote sensing images obtained from Google Earth and GaoFen-2 satellite while ensuring desirable speed performance. They first tackled three subtasks, key instance detection, environment analysis, and landscape analysis, and then integrated the results of each stage to generate captions. In contrast, Zhang et al. [59] used CNNs to detect the main objects in remote sensing images and used RNN to generate natural captions of the detected objects. Zhang et al. [60] proposed a visual alignment attention model that employs an elaborate visual alignment loss. This loss is determined by explicitly evaluating the feature similarity between the engaged image features and the corresponding word embedding vectors. To address the influence of nonvisual words in the training attention layer, they introduce a visual vocabulary that is eliminated from the sentence during the computation of the visual alignment loss. In addition, to bridge the semantic gap between low-level features and high-level semantic remote sensing images, Zhang et al. [18] utilized FCN to generate image features and attentional mechanisms to obtain intermediate vectors, which were then used as input to a long short-term memory (LSTM) decoder to produce captions of remote sensing images. Following this work, Li et al. [61] proposed a multilevel model of attention that contains three attentional structures: attention to different regions of an image, attention to different words, and visual and semantic attention. Zhang et al. [18] employed a transformer-based network for remote sensing image captioning. In particular, to handle limited training data, an auxiliary decoder is used for multilabel scene classification that exploits the conceptual similarity between the encoder and the image captions and the ability to highlight semantic classes, assisting the encoder during the training process. Wang et al. [62] proposed an interpretable framework for remote sensing image captioning, instead of using an interpretable word–sentence framework that consists of two networks that consist of a word extractor and a sentence generator. In addition, Li et al. [63] proposed a new truncated cross-entropy loss to solve the overfitting problem caused by a cross-entropy loss in remote sensing image captioning. To address the problem of ignoring domain knowledge in previous approaches, Sumbul et al. [24] proposed a fine-grained, structured attention-based model to exploit the structural features of semantic content in high-resolution remote sensing images. In addition, Zia et al. [25] first extracted detailed information from remote sensing images

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using a multiscale visual feature encoder and then refined the caption generation based on the extracted multiscale features using an adaptive multihead attention decoder.

Despite significant advances in multiscale approaches, they usually fail to adequately account for the interactions between visual and textual features at different scales, leading to inaccurate captioning. Inspired by [19] and [27], we propose a new ICNET framework based on transformer to address this problem. It acquires high-level concepts through CMN and integrates the contributions of these interactions through CIM. In addition, we introduce a GFE module in the extracted global representation to enhance the global discriminative representation of the model.

III. Methodology

The architecture of the proposed ICNET is shown in Fig. 1. The whole model mainly consists of a local and global feature extraction module, a CMN module, a GFE module, and a transformer with CIM. The ICNET mainly uses CMN to construct concepts and input them into the CIM to interact with word embeddings to improve the ability of the transformer to capture key targets. The transformer-based network mainly uses the VIT to extract the global features of the image and then uses the cross-attention mechanism for modeling relationships and generating final captions. In this section, global and local feature extraction, CMN, GFE, transformer with CIM, and model learning are described in detail.

A. Global and Local Feature Representation

In remote sensing images, objects usually display significant scale variations. Image representation, which encodes the valuable features and objects of a given remote sensing image into a high-level feature map, is crucial for understanding remote sensing images. High-quality image representation is beneficial for extracting more effective global and local features and further improving the caption performance [26].

Inspired by the above ideas, we obtain a comprehensive semantic representation of remote sensing images by extracting local features and global features using CNN and VIT, respectively. Local features capture the object information of remote sensing images, while global features summarize the relationship information of the complete remote sensing image.

For global features, they should have the characteristics of invariant translation and rotation. To achieve it, global feature $F_1 \in \mathbb{R}^{d_1}$ is extracted from the original RGB image $X \in \mathbb{R}^{W\times H\times 3}$ in a pretrained VIT. $W \times H$ is the image size, and 3 refers to the optical RGB channels. The process is denoted as

$$F_1 = \mathrm{VIT}(X). \tag{1}$$

For local features, different local features need to concentrate on different parts of remote sensing images. For this purpose, a local representation of the entire image is extracted by a pretrained ResNet152 image encoder. It yields four feature maps with different resolutions [12]. Since linguistic features are more discriminative than visual features, there is a large semantic gap between the two modal features, especially in the lower layers [53]. Therefore, three relatively deeper layers are chosen for a more comprehensive feature representation, denoted as $F_2, F_3, F_4 \in \mathbb{R}^{d_2}$, with specific sizes of 28×28, 14×14, and 7×7. This allows better interaction between visual and linguistic features. The convolutional process is denoted as

$$F_2, F_3, F_4 = \mathrm{ResNet}(X). \tag{2}$$

By employing these different level feature representations, we acquire rich local and global semantic information from remote sensing images. Moreover, the integration between vision and text enriches the embedding of information from both modalities.

B. Concept Mapping Network

The main purpose of the CMN is to construct semantic concepts for interacting with word embeddings in the decoder part of the transformer in order to align cross-modal features. To fully aggregate important information from local features at different scales, the proposed CMN employs attention mechanisms with dynamic channels and spatial to focus on critical semantics as well as structures. The weights are then mapped on the neighborhood feature maps to complement different levels of feature details. Finally, conceptual features are further refined through global average pooling operations. Specifically, the CNN-extracted multiscale local features F_2–F_4 are first processed through several small kernels of size 1. The process is formulated as

$$F_i' = \mathrm{Conv}(F_i), \quad i \in \{2, 3, 4\} \tag{3}$$

where $\mathrm{Conv}(\cdot)$ is the convolution operation.

Subsequently, the feature maps are resized to the same size $M \times K$ as F_3 with 512 channels. As shown in Fig. 1, assume that the shallow, intermediate, and deep feature maps are F_2'–F_4', respectively. F_2' contains abundant detail features, which is conducive to helping the model identify salient objects. To obtain the position weights of the hidden feature map, an average pooling operation on the channel dimension is employed, which is followed by a sigmoid operation. The position weight is then utilized in a matrix multiplication to fuse F_2' and F_3'. By doing so, the strategy achieves the information integration from the two feature maps according to position attention. It strengthens the representation of texture features in the fused feature maps. The process can be defined as

$$F_3'' = \frac{1}{C}\sum_{k=0}^{C} F_{2_{[i,j,k]}}' \odot F_3' \tag{4}$$

where C represents the channel number and $\odot$ denotes the matrix multiplication.

Consequently, deep feature maps F_4' can capture dense and large-scale targets. Pixel attention is generally applied to the finest granularity level of image. Therefore, pixel weights for N channels are generated at a deeper level using pixel attention, which focuses on the importance of each pixel and performs feature extraction by learning the importance of each pixel with respect to other pixels. It is conducive to

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learning the objects of remote sensing images in the decoder. For that reason, N soft attention modules are operated on F_4' to generate multiple critical semantic weights. It is the prerequisite to build N attention weighting maps with the spatial size of $M \times K$, which is denoted as $\gamma \in \mathbb{R}^{M\times K\times N}$. The weighting coefficient at the location of (i, j) in the nth attention map is calculated by

$$\gamma_{[i,j,n]} = \frac{1}{C}\sum_{k=0}^{C} F'_{4[i,j,k]} * w_{[n,k]} \tag{5}$$

where $\gamma_{[i,j,n]}$ is the intermediate weighting maps and $w_{[n,k]}$ projects C feature channels into N attention channels.

In order to be able to better align with the word embedding vectors in transformer's decoder and avoid introducing artificial parameters, we propose the position sharing mechanism, which uses the length of the word embedding vectors to align the attention weights N. It is then scaled by the SoftMax operation to obtain $\bar{\gamma}_{[i,j,n]}$ as follows:

$$\bar{\gamma}_{[i,j,n]} = \frac{\exp\left(\gamma_{[i,j,n]}\right)}{\sum_{i=0}^{M}\sum_{j=0}^{K}\exp\left(\gamma_{[i,j,n]}\right)}. \tag{6}$$

Finally, the global average pooling layer is used to aggregate the local feature maps into multiple concept feature vectors. The kth element in the nth vector of L is calculated by

$$L_{[n,k]} = \frac{1}{H\times W}\sum_{i=0}^{H}\sum_{j=0}^{W} F''_{3[i,j,k]} \otimes \bar{\gamma}_{[i,j,n]} \tag{7}$$

where $\otimes$ denotes the Hadamard product.

C. Global Feature Enhancement

To improve the discriminative abilities of global semantic information, this results in a more comprehensive and enriched feature representation. Further to filter irrelevant or weakly correlated content in the global feature F_1, we first use a 3×3 depthwise convolution block to convert it to a gated channel attention mechanism based on nearest neighbor features and to make its structure consistent with the design concept of gated channel attention. The process is formulated as

$$\hat{F}_1 = \mathrm{DwConv}(F_1). \tag{8}$$

Then, we bring an attention gate $G \in \mathbb{R}^{N\times d}$ based on the sigmoid activation function

$$G = \sigma\left(\hat{F}_1 W_g + b_g\right) \tag{9}$$

where $W_g \in \mathbb{R}^{d\times d}$ is the learnable parameter matrix, $b_g \in \mathbb{R}^d$ is the bias, and $\sigma(\cdot)$ is the sigmoid activation function. The attention gate G selectively emphasizes significant information within F_1, providing a more discriminative image representation. Based on this gate, the refined features $\tilde{F}_1 \in \mathbb{R}^{d_1}$ can be calculated as

$$\tilde{F}_1 = G \otimes F_1 \tag{10}$$

where $\otimes$ denotes the Hadamard product. Afterward, the refined features $\tilde{F}_1$ are fed into the transformer.

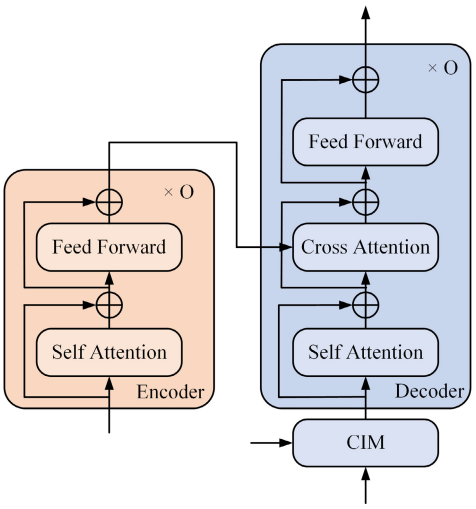


Fig. 2. Architecture of transformer with CIM. Based on the vanilla transformer [29], a conceptual interaction module is equipped at the head of the decoder.

D. Transformer With CIM

To take advantage of semantic concepts and global features, our transformer is composed of an encoder, a decoder, and the CIM, as shown in Fig. 2. The boosted global feature $\tilde{F}_1$ is fed into the transformer in a standard way, enhancing the model's ability to model global relationships. As for the semantic concepts, we enhance the correlation between semantic concepts and word embedding vectors through the proposed CIM, which is directly connected to the head of the decoder, and finally, together, they are used for word prediction.

1) CIM: Given the generated sentence denoted as S, get the feature vector E embedded by the word2vec [64] word embedding method and containing positional encoding, which is represented as follows:

$$E = \mathrm{Embeding(S)} + \mathrm{Postion}(N). \tag{11}$$

The process of generating language needs to be guided by semantic concepts for text generation. Therefore, based on attention mechanism, prefix causal language modeling (PCLM) is introduced, which aims at generating text by adapting given image prefixes, similar to the prefixes used in UniLM [65], yielding the following total attention weight:

$$\hat{X} = \mathrm{SoftMax}\left(\frac{(EW_q)(LW_k)^T}{\sqrt{d}}\right) \times (LW_v) \tag{12}$$

where E is the word vector; L is the concept vector; W_q, W_k, and W_v are the learnable parameter matrices; and d is the scaling factor.

The dot product is employed by the CIM to compute the similarity distribution between vectors E and L. The weighted sum $\hat{X}$ is further computed through the vector L with similarity metrics. As shown in Fig. 3, each text token embedding can handle all concepts and their preceding text token embeddings. PCLM is similar to traditional causal language modeling [66], where the model generates the text one token at a time, conditional on the preceding tokens. Note that the concepts

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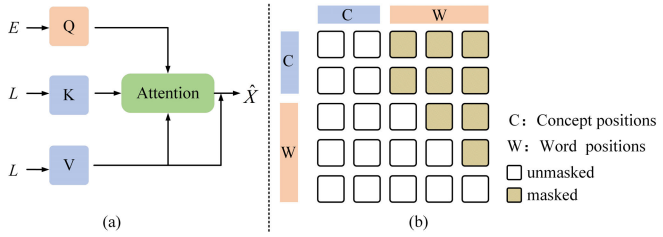


Fig. 3. Illustration of the CIM. (a) Interaction process between word embedding vectors and semantic concepts. (b) PCLM, where the masked squares represent masked text tokens that can attend to all semantic concepts and the preceding text tokens.

added as prefixes do not facilitate the computation of the PCLM loss, which is still essentially a sequence-to-sequence language modeling loss.

2) Transformer: As can be seen from Fig. 2, the standard transformer [29] is followed where the encoder is composed of a stack of O identical layers. Each encoding layer stacks self-attention blocks and a position-wise fully connected feed-forward network (FFN) in turn. A residual connection and a layer normalization are incorporated around the first and second sublayers, respectively. The inputs to each encoding layer include the output of the previous encoding layer. In particular, the inputs to the first layer use the output $\tilde{F}_1$ of the GFE module. In this way, the final output F_e of the encoder is obtained after O more coding layers.

The decoder is built with a stack of O identical decoding layers. Each decoder layer is composed of a masked self-attention, a cross-attention, and an FFN layer. We use a residual connection and layer normalization around each sublayer. Similar to the encoder, the inputs to each decoding layer include the output of the previous decoding layer and the output F_e of the encoder. For the first layer, the input is the output of the interactive concept network. Finally, the output F_d of the last decoding layer is regarded as the decoder output. Subsequently, F_d is fed into a fully connected layer and a SoftMax to calculate the probability over a vocabulary of possible words.

E. Training and Objectives

1) Training With Cross-Entropy Loss: Following a standard practice in image captioning [10], [11], we first train our model with a word-level cross-entropy loss (XE)

$$L_c(\theta) = -\sum_{t=1}^{T} \log\left(p_\theta\left(y_t \mid y_{[1,t-1]}\right)\right) \tag{13}$$

where $y_{[1,t-1]}$ represents a portion of the ground truth sequence, specifically the sequence of words from the start until the $(t-1)$th time step; T is the maximum time step; and θ denotes the variable parameter of the model.

2) CIDEr Score Optimization: However, recent work [23] has shown that cross-entropy training can lead to exposure bias, i.e., when the model is trained with ground truth words as input but tested with its own predictions as input. This discrepancy can cause the model to gradually accumulate errors, which can degrade its performance in sequential predictions. To this end, the self-critical sequence training reinforcement learning method [48] is applied to the image captioning task. It incorporates reasoning into the training process and introduces a sequence-level evaluation metric CIDEr as a reward function in the loss function. This approach helps to cope with the gap between the training and testing phases, while the model learns to generate words based on its previous predictions. The reward and loss functions are constructed as follows:

$$\nabla_\theta L_R(\theta) = -\frac{1}{k}\sum_{i=1}^{k}\left(\left(r\left(w^i\right) - b\right)\nabla_\theta \log p_\theta\left(w^i\right)\right) \tag{14}$$

where k is the number of samples; w^i represents the ith sentence sampled in the beam; $r(\cdot)$ indicates the reward function; and $b = 1/k\sum_{i=1}^{k} r(w^i)$ is the baseline reward, which is introduced to insure the stability of training.

TABLE I
STATISTICS OF THE DATASETS

Dataset	UCM-Captions	Sydney-Captions	NWPU-Captions
Image Number	2100	613	31500
Image size	256×256	500×500	256×256
Resolution	~0.3 m	0.5m	~30-0.2m
Sentences·image	5	5	5
Sentence number	10500	3065	157500
Scene classes	21	7	45

IV. Experimental Study

A. Experimental Settings and Datasets

To verify the performance of the proposed ICNET, we conduct experiments on three public datasets, including the Sydney-captions, UCM-captions, and NWPU-captions. Table I summarizes the details of the three datasets.

1) Sydney-Captions [57]: It is based on the Sydney dataset of remote sensing scene datasets [67]. The Sydney dataset contains 613 images categorized into seven scenes: residential, airport, grassland, river, ocean, industrial, and runway. The size of each image is 500 × 500 pixels with a pixel resolution of about 50 cm. All of these images were manually extracted from Sydney images downloaded from Google Earth with an original image size of 18 000 × 14 000 pixels.

2) UCM-Captions [57]: It is based on a remote sensing scene classification dataset of land use data from the University of California, Merced [68], which consists of 2100 images covering 21 typical land use scene categories, including agriculture, aircraft, baseball diamonds, beaches, buildings, brush, dense housing, forests, highways, golf courses, harbors, intersections, medium-density housing, mobile homes parks, overpasses, car parks, rivers, runways, sparse housing, oil storage tanks, and tennis courts. Each category has 100 images measuring 256 × 256 pixels with an RGB spatial pixel resolution of 30 cm. The UCM dataset was extracted from aerial orthophotos and downloaded from the USGS.

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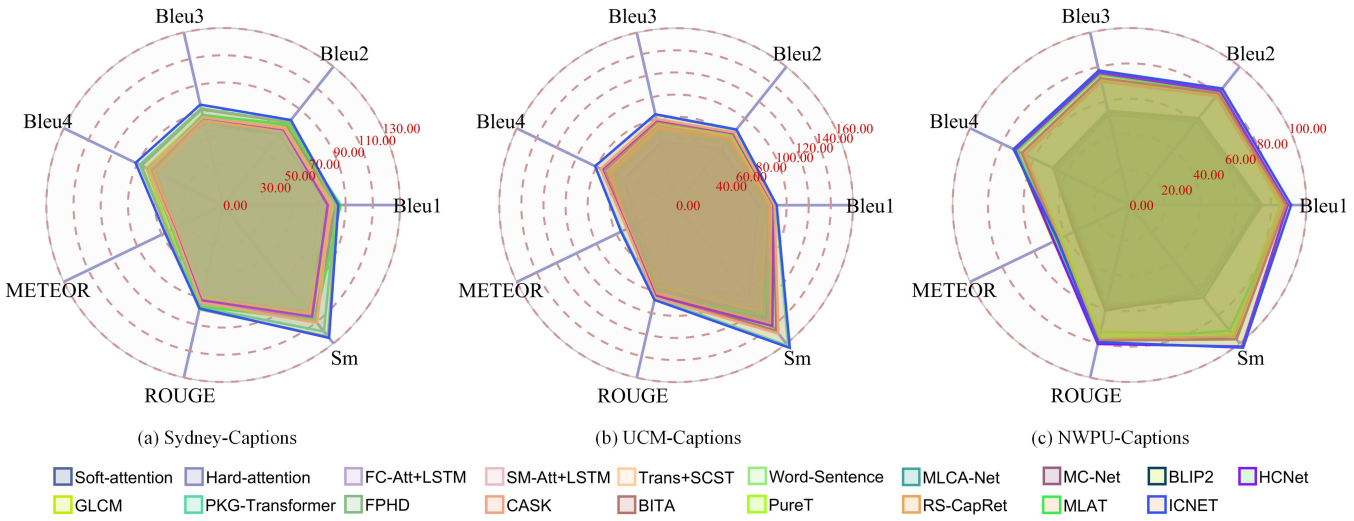


Fig. 4. Comparison results of state-of-the-art method on three datasets. (a) Syndey-captions. (b) UCM-captions. (c) NWPU-captions.

TABLE II
SPLITTING STRATEGY FOR TRAINING–VALIDATION–TESTING SETS

Dataset	Training	Validation	Test
UCM-Captions	1680	210	210
Sydney-Captions	497	58	58
NWPU-Captions	25200	3150	3150

3) NWPU-Captions [69]: It is built upon the remote sensing image classification dataset created by Northwestern Polytechnical University, which contains 31 500 aerial remote sensing images with RGB tri-frequency data and 157 500 sentences of image captioning. The spatial resolution of each image is 256 × 256 pixels, and the ground sampling distance ranges from 0.2 to 30 m. In addition, in order to enhance the semantic richness of the sentences, each image was accompanied by five different captions.

For each image in these three datasets, five sentences from different observers are provided. The division of these datasets into training, validation, and test sets is consistent with the original literature. They are all segmented in the same ratio of 80%/10%/10% on the training/validation/test sets, with the detailed sample counts as shown in Table II. In specific splits, some of the comparison methods do not use the default dataset splits but instead rely on random splits. For this reason, we statistics the splits of the comparison methods, where works [18], [28], [23], [74], [75] use random splits, works [19], [27], [62], [69] use standard splits, and works [22], [49], [76], [77], [78], [79], [80] do not explicitly state the splits. To ensure fairness with the comparison methods, inspired by [23] and [27], we conducted five experiments following the same ratio for each dataset. For each dataset, we conducted five experiments with five different random splits. In order to obtain more reliable results, the best and worst results were excluded, and the remaining results were averaged and the standard deviation calculated.

The proposed ICNET follows standard practice in remote sensing image captioning, and we adopt ResNet-152 pretrained on ImageNet to extract region features and adopt the pretrained CLIP image encoder with ViT-b/32 backbone to extract global features. By employing the ViT, the CLIP image embedding is able to encode visual tokens from all image patches. Moreover, the integration between vision and text enriches the embedding of information from both modalities. Following that a standard transformer framework with $N = 3$ identical layers is employed and head $= 8$ parallel attention layers are used in a multihead attention mechanism. For text representation, a start token ⟨bos⟩ and an end token ⟨eos⟩ are added before and after each ground truth sequence. In the training stage, the input to the decoder is the ground truth sequence. However, in the inference stage, the decoder only needs to input the start marker ⟨bos⟩. The word embedding dimension is configured as 512. The maximum sequence length for position encoding is 128 and the maximum output sequence length is 25. We set the batch size to 50 and optimize the model parameters using the Adam optimizer. To avoid overfitting, we discarded with a probability of 0.1 after each attention layer and FFN layer. The hidden dimension of each FFN layer was set to 2048. In addition, we defined "patience" to monitor the model's performance on the validation set and to decide whether the model needed to be improved. All experiments are conducted on NVIDIA Tesla V100-32 GB with Pytorch version 1.12.1.

During the cross-entropy training process, we adopt the learning rate scheduling strategy [29], which incorporates a warm-up operation comprising 10 000 iterations. We keep the model with the highest CIDEr score on the validation set as an initialization model for subsequent training phases or inference. When the patience value reaches 5, the training focus shifts to CIDEr optimization, and the learning rate is fixed at 5×10^{-6}. In both the optimization and decoding phases, we use a beam size of 5 to generate candidate sequences.

B. Evaluation Criteria

To assess the model's performance from various perspectives, we selected five commonly used evaluation metrics for

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image captioning tasks: BLEU [70], METEOR [71], ROUGE-L [72], CIDEr [73], SPICE, and S_m.

1) BLEU is a commonly used automatic evaluation metric to measure the similarity between generated text and reference text. It focuses more on the precision rate and calculates the score by calculating the proportion of n-grams (from single to multiple words) that appear in the reference text in the generated text and then combines exact and partial matches. Here, n-gram is a sequence of n continuous words with values ranging from 1 to 4.
2) METEOR combines exact and inexact matching criteria such as synonyms and stems. It uses external dictionaries to help identify synonyms and stems, allowing for a more comprehensive assessment of the quality of the generated text.
3) ROUGE is an evaluation metric to measure the recall between generated text and reference text. It evaluates the similarity between the generated text and the reference text by comparing the length of the longest common subsequence.
4) CIDEr is a metric for evaluating image captioning tasks that focuses on the diversity and consistency of generated captions. It calculates scores by comparing word frequency statistics between the generated text and multiple reference texts, emphasizing consistency across reference texts.
5) SPICE is a metric proposed for evaluating image captions. Unlike traditional n-gram-based evaluation methods that mainly evaluate by calculating the lexical overlap between the reference text and the generated text, SPICE uses semantic parse trees to measure the similarity between generated and reference texts, providing better semantic matching.
6) S_m is calculated as the arithmetic mean of BLEU-4, METEOR, ROUGE, and CIDEr metrics. It can evaluate the quality of generated captions comprehensively.

C. Compared Methods

To evaluate the effectiveness of the proposed ICNET, we compare our method with several state-of-the-art methods as follows.

Soft-attention and Hard-attention [74] introduce hard-attention and soft-attention mechanisms into the CNN-RNN framework, respectively.

FC-Att + LSTM and SM-Att + LSTM [18] integrate low-level features and high-level attribute features based on the attribute attention mechanism. The results are from [76].

Trans + SCST [75] proposes a new transformer structure. It added a dropout layer, residual connection, and adaptive feature fusion on the transformer. Reinforcement learning is used to improve the actual performance of the model.

Word-Sentence [62] is a typical encoder–decoder framework consisting of a word extractor and a sentence generator. This method aims to extract words from remote sensing images and form coherent sentences.

MLCA-Net [69] aggregates features explored at different scales and potential contexts through multilevel and context-attentive modules and produces captions through LSTM.

MC-Net [76] contains an encoder enhanced with a multiscale extraction module, a feature fusion module, and a fine-tuned LSTM equipped with a visual–text alignment module.

GLCM [27] is an attention-based global–local captioning model. Global and local features are extracted by CNN for sentence generation.

PKG-Transformer [23] first enriches the object and scene features by leveraging the object–object and scene–scene relationship. Then, it integrates the scene–object relationship into the transformer encoder as prior knowledge.

FPHD [77] designs a deformable transformer by separating foreground and background based on hierarchical image information and learns multiscale features from foreground and background through interaction.

BITA [78] utilizes a lightweight interactive Fourier transform to align remote sensing image–text features and a two-level visual-language pretraining approach to guide interactive image–text alignment.

CASK [79] captures semantic concepts through a semantic concept extractor. Joint visual-semantic co-attention is incorporated into the consensus development module for learning cross-modal consensus-aware knowledge.

PureT [49] employs a swin transformer to extract features from an image; a refinement encoder is then used to capture the relationships between features, and a decoder decodes the refined features verbatim as captions. The results are from [19].

RS-CapRet [22] uses CLIP encoding for captions and generates captions and text–image retrieval by using a simple fusion network and then fine-tuning the LLamaV2-7B model.

BLIP-2 [80] is a visual-language pretraining approach that employs a Q-Former structure inspired by the BERT base for multimodal fusion.

HCNet [19] is a method that uses CNN for hierarchical visual feature extraction and LSTM for language generation.

MLAT [28] utilizes LSTM to fuse features from different layers in ResNet-50 to extract multiscale information. The captions are then generated by a multilayer aggregation transformer.

D. Comparison With Existing Methods

This section provides a selected comparison of the proposed ICNET with several state-of-the-art methods, including attention- and transformer-based methods. The results are presented in Tables III–V. The experimental results showed that the transformer-based method performed better than the other methods to a certain extent in Fig. 4. The best experimental metrics are obtained using the proposed model for most metrics. This was because our model not only used the semantic concepts and retained the global helpful information for subsequent text generation but also retained the features of the salient areas or objects in the image, which were improved using the mapping local features. This improves the ability of the model to perceive features in an object's neighborhood. During text generation, the decoder is directly connected to

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TABLE III

COMPARISON OF SOME STATE-OF-THE-ART METHODS ON SYDNEY-CAPTIONS. ALL RESULTS ARE REPORTED AS PERCENTAGE (%). THE SYMBOL "-" INDICATES THAT THE RESULT IS NOT REPORTED BY THIS ARTICLE

Method	Bleu1	Bleu2	Bleu3	Bleu4	METEOR	ROUGE	CIDEr	SPICE	S_m
Soft-attention [74]	73.22	66.74	62.23	58.20	39.42	71.27	249.93	-	104.71
Hard-attention [74]	75.91	66.10	58.89	52.58	38.98	71.89	218.19	-	95.41
FC-Att+LSTM [18]	73.80	64.40	57.00	50.90	36.40	66.90	224.20	39.50	94.6
SM-Att+LSTM [18]	74.30	63.40	58.60	51.80	36.40	67.70	234.00	39.80	97.48
Trans+SCST [75]	80.16	70.95	63.14	57.24	41.83	72.62	253.92	-	106.40
Word-Sentence [62]	78.91	70.94	63.17	56.25	41.81	69.22	204.11	-	92.85
MLCA-Net [69]	83.10	74.20	65.90	58.00	39.00	71.10	232.40	40.90	100.13
MC-Net [76]	83.40	75.00	67.80	60.70	40.60	73.90	256.40	45.60	107.90
GLCM [27]	80.41	73.05	67.45	62.59	44.21	69.65	243.37	-	104.96
PKG-Transformer [23]	83.17±1.02	77.83±2.31	72.84±1.88	68.24±1.39	45.28±0.86	77.06±1.54	284.76±10.84	44.05±1.05	118.83±2.51
FPHD [77]	83.73	77.71	71.98	66.59	45.48	**78.60**	**303.69**	**48.39**	123.59
MLAT [28]	**86.29**	75.85	67.48	60.33	41.51	76.11	255.75	39.04	108.43
RS-CapRet [22]	78.70	70.00	62.80	56.40	38.80	70.70	239.20	42.30	101.28
HCNet [19]	76.86	71.09	65.73	61.02	39.80	71.72	247.14	-	104.92
CASK [79]	79.08	72.00	66.05	60.88	40.31	73.54	267.88	46.37	110.65
ICNET	84.68±0.68	**79.85±1.77**	**75.61±3.47**	**71.50±4.57**	**47.40±4.59**	77.91±1.3	302.58±8.78	47.28±3.94	**124.85±2.04**

TABLE IV

COMPARISON OF SOME STATE-OF-THE-ART METHODS ON UCM-CAPTIONS. ALL RESULTS ARE REPORTED AS PERCENTAGE (%). THE SYMBOL "-" INDICATES THAT THE RESULT IS NOT REPORTED BY THIS ARTICLE

Method	Bleu1	Bleu2	Bleu3	Bleu4	METEOR	ROUGE	CIDEr	SPICE	S_m
Soft-attention [74]	74.54	65.45	58.55	52.50	38.86	72.37	261.24	-	106.24
Hard-attention [74]	81.57	73.12	67.02	61.82	42.63	76.98	299.47	-	120.22
FC-Att+LSTM [18]	81.00	73.30	67.30	61.90	42.80	76.70	337.00	48.70	129.6
SM-Att+LSTM [18]	81.20	74.20	68.10	63.00	43.50	77.90	338.60	48.80	130.75
Trans+SCST [75]	83.82	79.01	74.43	70.11	44.64	77.63	356.53	-	137.23
Word-Sentence [62]	79.31	72.37	66.71	62.02	43.95	71.32	278.71	-	114.00
MLCA-Net [69]	82.60	77.00	71.70	66.80	43.50	77.20	324.00	47.30	127.88
MC-Net [76]	84.50	78.40	73.20	67.90	44.90	78.60	335.50	52.00	131.73
BLIP-2 [80]	88.04	82.23	76.89	71.84	47.32	83.36	380.31	54.78	145.71
GLCM [27]	81.82	75.40	69.86	64.68	46.19	75.24	302.79	-	122.23
PKG-Transformer [23]	90.48±1.15	87.04±1.09	84.10±1.33	**81.39±1.67**	54.66±2.05	86.57±2.00	427.49±12.37	57.01±2.40	162.53±3.52
PureT [49]	85.73	80.20	75.62	71.29	46.86	82.01	349.00	47.94	137.29
FPHD [77]	82.30	77.00	72.28	67.92	44.39	78.39	346.29	48.25	134.25
MLAT [28]	89.96	83.19	77.52	72.02	48.27	84.09	305.01	25.93	127.35
BITA [78]	88.80	83.12	77.30	71.87	46.88	83.76	384.50	54.88	146.75
RS-CapRet [22]	84.30	77.90	72.20	67.00	47.20	81.70	354.80	52.50	137.68
HCNet [19]	88.26	83.35	78.85	74.49	48.65	83.91	351.83	-	139.72
CASK [79]	89.00	84.16	79.87	75.75	49.31	85.78	383.14	52.27	148.50
ICNET	**91.02±0.65**	**87.52±0.89**	**84.31±0.96**	81.35±1.00	**55.25±0.14**	**87.85±0.15**	**435.66±3.68**	**58.18±1.23**	**165.02±1.16**

TABLE V

COMPARISON OF SOME STATE-OF-THE-ART METHODS ON NWPU-CAPTIONS. ALL RESULTS ARE REPORTED AS PERCENTAGE (%). THE SYMBOL "-" INDICATES THAT THE RESULT IS NOT REPORTED BY THIS ARTICLE

Method	Bleu1	Bleu2	Bleu3	Bleu4	METEOR	ROUGE	CIDEr	SPICE	S_m
Soft-attention [74]	73.10	60.90	52.50	46.20	33.90	59.90	113.60	-	63.40
Hard-attention [74]	73.30	61.00	52.70	46.40	34.00	60.00	110.30	-	62.68
FC-Att+LSTM [18]	73.80	61.30	53.40	47.10	34.10	60.20	113.50	28.20	63.73
SM-Att+LSTM [18]	73.90	61.60	53.40	46.90	33.80	59.50	113.70	27.90	63.48
MLCA-Net [69]	74.50	62.40	54.10	47.80	33.70	60.10	126.40	28.50	67.00
MC-Net [76]	74.10	62.60	54.40	47.80	34.70	61.10	115.90	28.90	64.88
BLIP-2 [80]	87.78	80.24	73.51	67.62	45.01	78.44	193.86	33.28	96.23
GLCM [27]	85.37	80.44	75.93	72.03	**47.77**	73.56	179.32	-	93.17
FPHD [77]	75.15	62.91	54.57	48.28	31.87	58.58	120.71	26.78	64.86
MLAT [28]	89.16	81.78	75.65	70.51	45.93	78.72	192.83	8.20	90.78
PureT [49]	88.82	80.31	73.30	67.50	42.32	75.84	195.12	27.43	95.20
RS-CapRet [22]	87.10	78.70	71.70	65.60	43.60	77.60	192.90	31.10	94.93
HCNet [19]	89.54	82.51	76.58	71.68	47.42	**80.53**	209.27	-	102.22
BITA [78]	88.54	80.70	73.56	67.60	45.27	78.53	197.04	**33.65**	97.11
ICNET	**91.39±0.23**	**84.18±0.28**	**77.97±0.42**	**72.79±0.69**	45.02±0.82	79.47±0.44	**215.02±0.78**	28.43±0.42	**103.08±0.33**

concepts of all types, unlike the conventional transformer in which the decoder only attends to the last-layer outputs of the encoder. The configuration used in the proposed method allows different types of features to participate in text generation, providing both global and local area salient information. This demonstrates the superiority of the ICNET.

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TABLE VI
RESULT OF ABLATION STUDY ON SYDNEY-CAPTIONS. ALL RESULTS ARE REPORTED AS PERCENTAGE (%)

Method	Bleu1	Bleu2	Bleu3	Bleu4	METEOR	ROUGE	CIDEr	SPICE	S_m
Baseline	79.61±0.09	71.26±0.55	65.41±1.07	60.57±1.49	40.49±2.83	70.99±3.43	268.18±25.79	41.79±5.14	110.06±8.35
Model-1	84.29±0.29	78.87±0.77	74.06±1.36	69.66±1.63	43.46±3.87	75.89±2.85	288.30±24.27	40.78±8.05	119.33±8.06
Model-2	83.99±0.90	77.68±0.48	72.94±1.09	68.86±1.84	45.53±2.61	77.63±3.14	299.52±15.21	45.84±4.47	122.89±5.60
Model-3	83.71±0.36	77.18±1.73	72.08±2.62	67.77±3.28	44.03±3.51	76.30±3.38	**307.88±11.65**	42.63±2.91	123.99±3.64
ICNET	**84.68±0.68**	**79.85±1.77**	**75.61±3.47**	**71.50±4.57**	**47.40±4.59**	**77.91±1.3**	302.58±8.78	**47.28±3.94**	**124.85±2.04**

TABLE VII
RESULT OF ABLATION STUDY ON UCM-CAPTIONS. ALL RESULTS ARE REPORTED AS PERCENTAGE (%)

Method	Bleu1	Bleu2	Bleu3	Bleu4	METEOR	ROUGE	CIDEr	SPICE	S_m
Baseline	87.58±0.05	83.34±0.39	79.62±0.34	76.55±0.51	51.66±1.91	84.71±2.80	384.23±16.00	50.67±0.18	149.29±5.22
Model-1	88.84±0.23	84.50±0.10	81.03±0.32	78.17±0.87	52.51±3.46	84.92±1.19	399.81±28.10	51.05±1.16	153.85±8.40
Model-2	89.08±0.54	85.56±0.47	82.26±0.93	79.36±1.26	53.27±0.44	86.03±1.14	399.68±4.12	53.26±1.55	154.59±1.19
Model-3	89.31±1.02	85.79±1.68	82.83±2.12	80.24±2.53	52.71±0.26	85.75±0.97	423.73±16.97	54.42±1.18	160.61±5.15
ICNET	**91.02±0.65**	**87.52±0.89**	**84.31±0.96**	**81.35±1.00**	**55.25±0.14**	**87.85±0.15**	**435.66±3.68**	**58.18±1.23**	**165.02±1.16**

TABLE VIII
RESULT OF ABLATION STUDY ON NWPU-CAPTIONS. ALL RESULTS ARE REPORTED AS PERCENTAGE (%)

Method	Bleu1	Bleu2	Bleu3	Bleu4	METEOR	ROUGE	CIDEr	SPICE	S_m
Baseline	88.32±0.61	80.83±0.55	74.33±1.11	68.83±1.46	40.73±1.16	76.38±0.74	209.17±4.69	25.82±0.05	98.78±1.51
Model-1	88.48±1.04	80.85±0.88	74.38±0.70	68.93±0.65	41.33±0.18	76.39±0.30	207.38±1.47	25.88±0.14	98.51±0.57
Model-2	89.61±0.44	82.04±0.34	75.52±0.22	70.05±0.16	41.92±0.41	76.65±0.06	209.73±0.33	25.69±0.23	99.59±0.24
Model-3	90.61±0.43	83.26±0.79	76.86±0.96	71.51±1.02	42.79±1.36	77.58±1.31	211.26±2.78	26.69±1.79	100.79±1.56
ICNET	**91.39±0.23**	**84.18±0.28**	**77.97±0.42**	**72.79±0.69**	**45.02±0.82**	**79.47±0.44**	**215.02±0.78**	**28.43±0.42**	**103.08±0.33**

E. Ablation Study

The contributions of each module to the overall performance of ICNET are evaluated through the ablation study. The proposed ICNET with only the VIT and transformer is used as the baseline model. The three modules are added one by one to construct the ablation models. These models are as follows: Model-1 is baseline + GFE module, Model-2 is baseline + CNN + CIM module, Model-3 is baseline + CMN + CIM module, and ICNET is baseline + GFE + CMN. Eight overall assessment criteria (i.e., Bleu1, Bleu2, Bleu3, Bleu4, METEOR, ROUGE, CIDEr, and S_m) are used to evaluate the performance of three test image caption datasets, which are summarized in Tables VI–VIII.

1) CMN: We explored the role of CMN in our model through baseline, Model-2, and Model-3. The CMN module is used to convert local features extracted by CNN into semantic concepts for interaction with word embeddings. For Model-2, we directly apply the CNN-extracted features for interaction using global average pooling. The results of the test set on three sets of data show that Model-3 outperforms Model-2, while Model-2 also shows significant improvement compared to the baseline due to the introduction of local features. This also verifies that the utilization of local features with global features is beneficial to improve the performance of the model.

2) GFE: Analogously, the impact of the GFE module is explored on the three datasets. From the comparison between the baseline and Model-1, we can visualize in Fig. 5 that the implementation of the GFE module improves the model performance by filtering the irrelevant or weakly relevant content models in the global features. For example, the phrase "a lawn beside" was filtered in the results for the industrial area scene corresponding to the Sydney-captions dataset. Among other things, attentional gating selectively emphasizes important information in features, providing a more robust global representation with discriminative properties.

3) Transformer With CIM: The effectiveness of the CIM module was validated on three datasets. Tables IX–XI perform in contrast to the different interactions. It can be seen that the CIM module has a better performance compared to the traditional feature addition and multiplication. It can also be seen in Fig. 6 that the three interaction methods show a similar distribution in the comparison of the eight quantitative metrics. The proposed method, on the other hand, possesses better metrics. On the Sydney-captions dataset, the improvements in the S_m score are achieved by 9.58% (over add) and 5.57% (over dot). On the UCM-captions dataset, the improvements in the S_m score are achieved by 7.55% (over add) and 3.95% (over dot). On the NWPU-captions dataset, the improvements in the S_m score are achieved by 3.79% (over add) and 2.68% (over dot). Finally, ICNET is building on the baseline and contains all the components mentioned above. It can illustrate the capabilities of CMN and CIM compared to Model-1 and the advantages of GFE compared to Model-3. The integration of all components improves the ability of the model to resist interference factors and enhances the representation between global and local components of the model. In addition, the strategy enhances the interaction between image features and word features with good alignment of representation capability.

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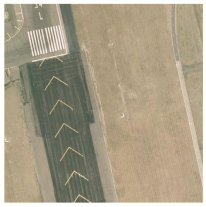
Sydney-Captions



GT1: A crossroad in the middle of the industrial area.
GT2: Many white buildings in the industrial area while a crossroad in the middle.
GT3: A crossroad is in the middle of the industrial area with many white houses densely arranged.
GT4:Many houses are densely arranged while a crossroad in the middle.
GT5:An industrial area with many white buildings densely arranged while a crossroad in the middle.
Baseline: An industrial area with many white buildings densely arranged while a lawn beside.
Model-1: An industrial area with many white buildings and some roads go through.
GLCM: An industrial area with many white buildings densely arranged.
ICNET: An industrial area with some white buildings densely arranged while some roads go through.



GT1: A residential area with many houses arranged neatly.
GT2: Many houses arranged densely and divided into rectangles by some roads.
GT3:This is a residential area with many houses neatly arranged.
GT4:Lots of houses arranged neatly in the residential area and divided into rectangles by some roads.
GT5:There are many houses arranged neatly and divided into rectangles by some roads.
Baseline: A residential area with many houses arranged neatly and divided into rectangles by some roads.
Model-1: A residential area with houses arranged neatly and some roads go through this area
GLCM: A residential area with many houses arranged neatly and divided by some roads.
ICNET: A residential area with houses arranged neatly and some roads go through this area.



GT1: A straight runway with some white mark lines and arrows on.
GT2: There are some white mark lines and arrows on the straight runway.
GT3:A straight runway with some mark lines and arrows on while some lawns beside.
GT4:There are some lawns beside the width straight runway.
GT5:A straight runway with some white mark lines and arrows while some lawns beside.
Baseline: There are some white marking lines on it while some lawns beside.
Model-1: A straight runway with many marking lines on it while a lawns beside.
GLCM: There are some white marking lines on the runways.
ICNET: There are some marking lines on the runways while some lawns beside.

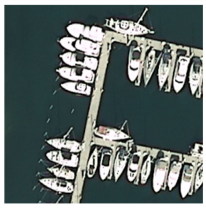
UCM-Captions



GT1: Lots of cars are very closed to each other in the parking lot.
GT2: Lots of cars parked neatly in the parking lot.
GT3:Many cars are very closed to each other in the parking lot.
GT4:It is a parking lot with many cars densely arranged.
GT5:Many cars densely arranged in the parking lot.
Baseline: Many cars densely arranged in the parking lot.
Model-1: Lots of cars parked in lines in the parking lot.
GLCM: Many cars densely arranged in the parking lot.
ICNET: Lots of cars are very closed to each other in the parking lot.



GT1: An intersection with no cars or pedestrians.
GT2: An intersection with some plants at the corners.
GT3:A desolate intersection only with some plants at the corners.
GT4:A desolate intersection with no cars or pedestrians.
GT5:An intersection with two roads vertical to each other.
Baseline: An intersection with two roads are not vertical to each other.
Model-1: An intersection with two roads vertical to each other.
GLCM: An intersection with some houses and plants at the corners.
ICNET: An intersection with two roads vertical to each other.



GT1: Lots of boats docked at the harbor and the water is deep blue.
GT2: Lots of boats docked neatly at the harbor.
GT3:Many boats docked neatly at the harbor and the water is deep blue.
GT4:Many boats docked neatly at the harbor and only a few positions are free.
GT5:Lots of boats docked neatly at the harbor and only a few positions are free.
Baseline: There is a piece of farmland.
Model-1: Many boats docked neatly at the harbor and the water.
GLCM: Lots of boats docked at the harbor and only a few position are free.
ICNET: Many boats docked neatly at the harbor and the water is deep blue.

NWPU-Captions



GT1: The two cargo ships are docked at the container terminal and one of the ships is full of containers.
GT2: Two white general cargo ships docked at the dock, With large swaths of cargo on the dock.
GT3:There are two ships in the blue water on the shore.
GT4:There are many containers in the dock beside the ships.
GT5:There are two ships on the harbor with many containers.
Baseline: There is a ship in the black water on the shore.
Model-1: There are two ships in the green water on the shore.
GLCM: There is four ships in the blue water on the shore.
ICNET: There are two ships in the blue water on the shore.



GT1: Here are two white storage tanks of the same size.
GT2: There are many different sized storage tanks and many buildings in the factory.
GT3:There are many storage tanks on the bare land next to buildings.
GT4:There are many buildings of different sizes near the storage tanks.
GT5:There are many white storage tanks and buildings.
Baseline: Many houses of different shapes are in the industrial area.
Model-1: There are some blue buildings on the industrial area.
GLCM: There are many buildings and some green trees near the industrial area.
ICNET: The thermal power station is on the grass next to some buildings.



GT1: This wetland is totally covered with vegetation.
GT2: The blue lake in a wetland is covered by large swaths of dense vegetation.
GT3:The wetland consists of bare land, Green plants and water.
GT4:Many waters of different sizes are surrounded by wetlands.
GT5:There are many green areas and water areas on the wetland.
Baseline: Many waters of different sizes are surrounded by wetlands.
Model-1: There are many different areas of water surrounded by wetlands.
GLCM: There are many water areas and some bare land on the wetland.
ICNET: The wetland consists of bare land green plants and water.

Fig. 5. Example sentences generated by baseline, Model-1, GLCM, and ICNET methods on three datasets. The GT denotes the ground truth captions. The words in red incorrectly match the corresponding image, while some crucial objects and relationship words are highlighted in green.

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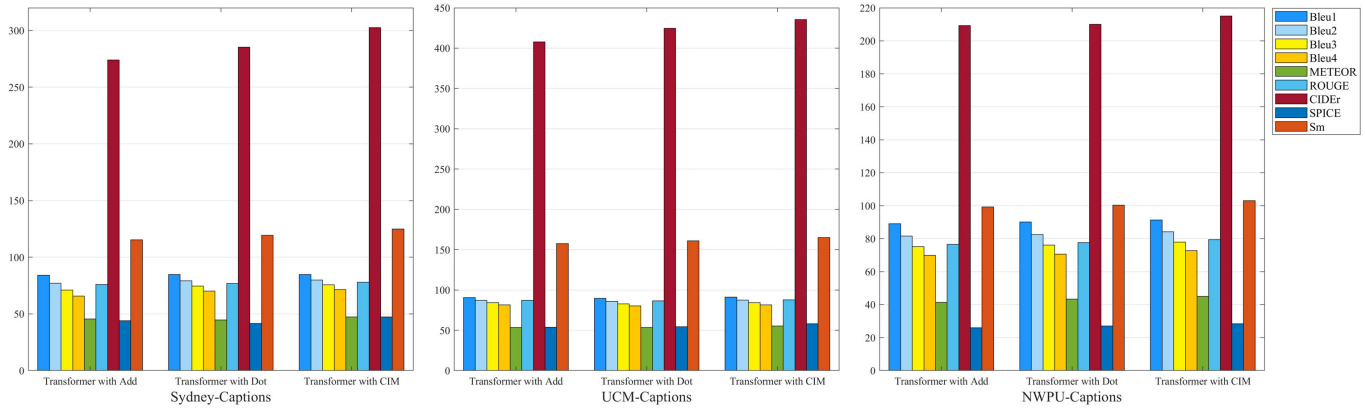


Fig. 6. Comparison results of ICNET with different interactive modules on three datasets.

TABLE IX

COMPARISON OF DIFFERENT INTERACTION MODULES ON SYDNEY-CAPTIONS. ALL RESULTS ARE REPORTED AS PERCENTAGE (%)

Method	Bleu1	Bleu2	Bleu3	Bleu4	METEOR	ROUGE	CIDEr	SPICE	S_m
Transformer with Add	84.13±0.58	77.06±0.49	70.96±1.40	65.61±1.79	45.57±2.16	75.95±1.21	273.94±20.94	43.97±1.88	115.27±5.07
Transformer with Dot	**84.74±0.77**	79.24±1.94	74.56±2.57	70.19±2.86	44.76±1.98	76.89±1.06	285.29±19.45	41.62±4.52	119.28±3.98
Transformer with CIM	84.68±0.68	**79.85±1.77**	**75.61±3.47**	**71.50±4.57**	**47.40±4.59**	**77.91±1.30**	**302.58±8.78**	**47.28±3.94**	**124.85±2.04**

TABLE X

COMPARISON OF DIFFERENT INTERACTION MODULES ON UCM-CAPTIONS. ALL RESULTS ARE REPORTED AS PERCENTAGE (%)

Method	Bleu1	Bleu2	Bleu3	Bleu4	METEOR	ROUGE	CIDEr	SPICE	S_m
Transformer with Add	90.63±0.49	87.21±0.50	84.19±0.48	**81.38±0.24**	53.55±0.76	87.21±0.97	407.72±24.50	53.66±0.76	157.47±6.58
Transformer with Dot	89.47±0.47	85.92±0.66	82.79±0.86	80.15±0.88	53.29±1.93	86.37±1.08	424.48±15.16	54.27±1.97	161.07±3.15
Transformer with CIM	**91.02±0.65**	**87.52±0.89**	**84.31±0.96**	81.35±1.00	**55.25±0.14**	**87.85±0.15**	**435.66±3.68**	**58.18±1.23**	**165.02±1.16**

TABLE XI

COMPARISON OF DIFFERENT INTERACTION MODULES ON NWPU-CAPTIONS. ALL RESULTS ARE REPORTED AS PERCENTAGE (%)

Method	Bleu1	Bleu2	Bleu3	Bleu4	METEOR	ROUGE	CIDEr	SPICE	S_m
Transformer with Add	89.07±1.04	81.62±0.77	75.25±0.50	69.85±0.43	41.41±2.36	76.58±1.17	209.33±2.17	25.97±1.42	99.29±1.04
Transformer with Dot	90.11±0.13	82.55±0.16	76.03±0.14	70.58±0.11	43.39±0.61	77.64±0.25	209.98±0.17	26.99±0.73	100.40±0.19
Transformer with CIM	**91.39±0.23**	**84.18±0.28**	**77.97±0.42**	**72.79±0.69**	**45.02±0.82**	**79.47±0.44**	**215.02±0.78**	**28.43±0.42**	**103.08±0.33**

F. Qualitative Analysis

1) Caption Generation Results: As shown in Fig. 5, we extracted several caption results generated by our model from three datasets: Sydney-captions, UCM-captions, and NWPU-captions. For comparison, we show nine captioning results: the results of our model, the results of the state-of-the-art method GLCM; the results of baseline; the results of Model-1, the underlying component that comprises our model; and five ground-truth labeling results such as GT1–GT5. As can be seen in Fig. 5, our method generates captions that are more accurate in describing objects with lower error rates than the GLCM method.

Specifically, notice that in the 1st image of the Sydney-captions dataset in Fig. 5, the foreground (road) is hidden in the background (industrial area). In this case, generating sentences containing roads by the benchmark method is very challenging and lacks interpretability. In fact, the captions generated by the benchmark method do not contain road information. However, our model can efficiently extract this hidden foreground and reflect it in the generated sentences. Also, the adequacy of the sentences meets the requirements. All these sentence improvements are attributed to the fact that our CMN and global feature boosting module can reduce the interference of external noise during the feature interaction learning process. A similar situation exists in the samples shown in the UCM dataset in Fig. 5. Specifically, the third sample "harbor" is surrounded by many ships, which is difficult to recognize without a priori. As a result, the benchmark method lacks this object in the generated sentence. In the 1st sample of the NWPU-captions dataset in Fig. 5, the number of boats is recognized because it is close to the target about "shore" and the target itself is narrow. Our proposed CMN, which also incorporates the idea of multiscale, can recognize the number of boats very well. This also demonstrates the strong generalization ability of the proposed method. Other samples also demonstrate the superiority of the model. In the second sample, the missing "electric power heating plant"

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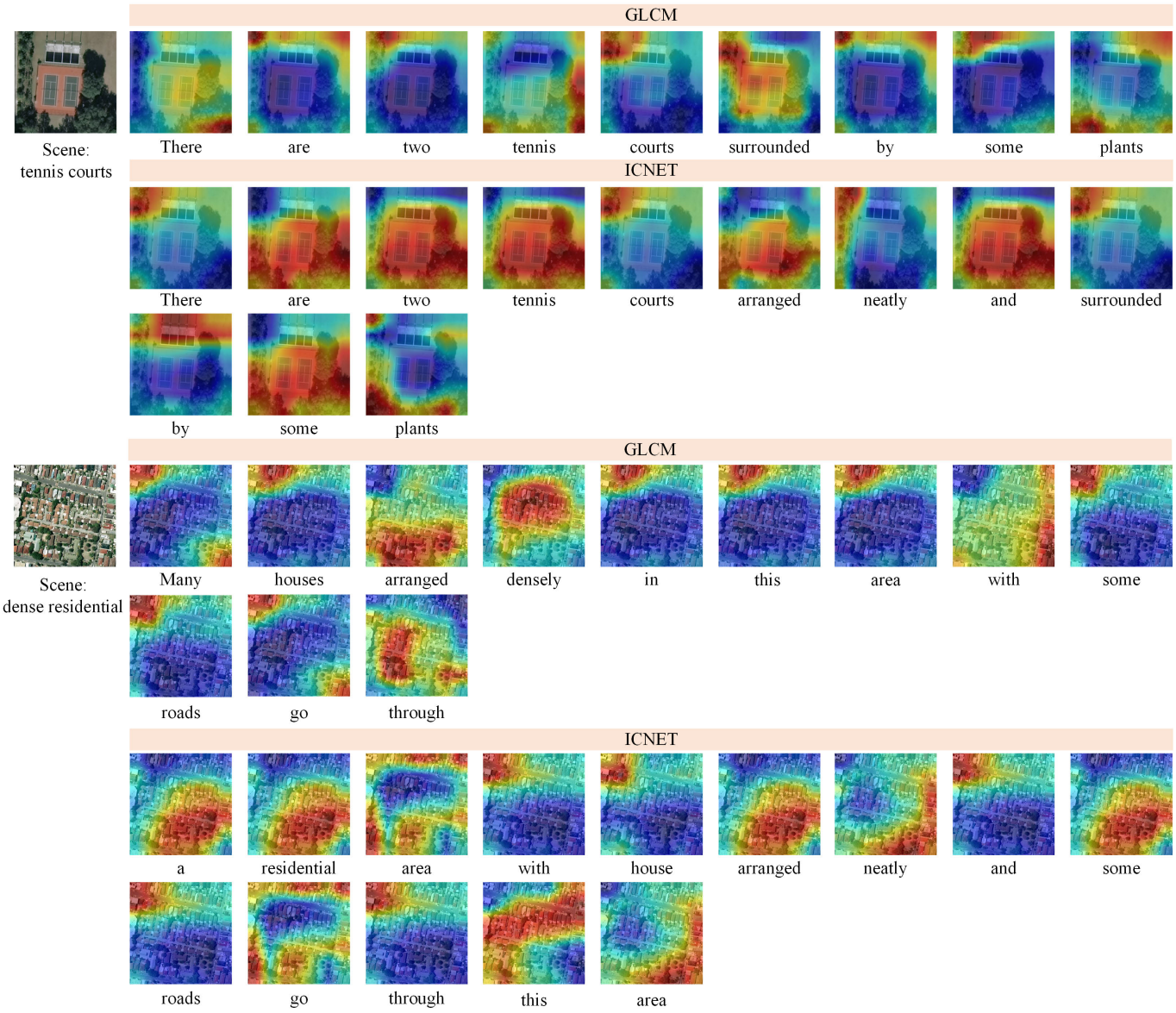


Fig. 7. Visualization of attended image regions along the caption generation processes for ICNET and GLCM. The red indicates areas with high attention.

is detected by our method. Our model also gives surprising results in captioning the wetland consisting of green plants and water bodies shown in the third sample.

2) Visualization of the Attention: Benefiting from the attention layer in the CMN, the semantic concepts that best match the word vectors can be visualized from the attention layer. Fig. 7 shows the comparison of the proposed ICNET and GLCM methods for attention visualization. Compared to the GLCM method, most of the activation regions are more plausible, although not all words are located exactly in the image. It is also interesting to note that some consecutive words respond to the same local features. For example, in the tennis scene and the residential scene, "two tennis" and "a residential" are clustered into two corresponding local features. It reflects the hierarchical structure from words to phrases to sentences in natural language.

3) Comparison of Algorithmic Complexity: This section provides a comparison of the computational complexity of the proposed method with several related methods, especially the number of parameters and FLOPs. Table XII shows the comparison of the specific number of parameters and the FLOPs. Compared to GLCM, ICNET is proposed with a larger number of parameters. This is because ICNET extracts more comprehensive features based on CNN and VIT, but

TABLE XII

COMPARISON BETWEEN THE MLAT, PKG-TRANSFORMER, GLCM, BASELINE, AND THE PROPOSED ICNET ON THE NUMBER OF PARAMETERS AND FLOPS. THE SYMBOL "-" INDICATES THAT THE RESULT IS NOT REPORTED BY THIS ARTICLE

Method	Parameters	FLOPs
MLAT [28]	149.77M	-
PKG-Transformer [23]	31.94M	1.58G
GLCM [27]	13.49M	1.89G
Baseline	22.42M	0.83G
ICNET	24.18M	1.04G

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ICNET can achieve better captions. Compared to MLAT and PKG-Transformer methods, the proposed ICNET has the smallest number of parameters and also achieves good results. Ultimately, compared to the baseline, ICNET performance has a more significant improvement with a similar number of parameters.

V. Conclusion

This article proposes a novel remote sensing image captioning method, ICNET, based on multilevel feature aggregation and cross-modal feature interaction. ICNET aims to utilize multiscale local features and global features to guide the model to produce accurate objects and relationship captions. Based on the transformer architecture, a CMN, a CIM, and a GFE are introduced. To enhance the local feature representation capability, the CMN is proposed to efficiently aggregate local features at different scales to generate high-level semantic concepts. The semantic concepts are further modeled to interact with word vectors through the CIM, to integrate the relevant feature vectors in the visual feature mapping into multiple relatively independent word features, and to narrow the gap between visual features and semantic concepts. To enhance the global discriminative capability of the model, more comprehensive and discriminative feature representations are obtained through the GFE, which are injected into the model in a cross-attention way, ultimately yielding robust and well-performing captions. The comparison analysis with the state-of-the-art models shows outperforming captioning results on three remote sensing image captioning datasets. Furthermore, the contribution of each module is analyzed through an ablation study.

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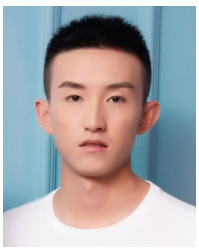
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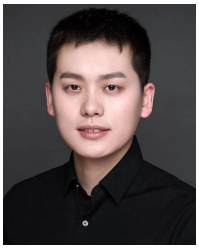
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